Sentinel-1 SAR Oil Spill Predictor

Simple Terms & How They Work in Your Project

A beginner-friendly study guide based on the technical documentation you provided. It explains what each major term means, what it does, and where it appears in your pipeline.

Big picture: Sentinel-1 image → prepare data → ResNet34 + U-Net → predict a class for each pixel → calculate loss → train repeatedly → evaluate on unseen images → clean the prediction → convert it into geographic polygons → GeoJSON.

Important: when the documentation does not establish why a particular number was chosen, this guide labels that as an implementation value rather than inventing a reason.

1. Sentinel-1

Simple meaning: What it is: A satellite mission whose radar imagery is used as the input to your project.

In your project: In your project: Sentinel-1 supplies SAR imagery rather than ordinary RGB photographs. The model learns patterns in this radar data to identify oil-spill-related regions.

2. SAR — Synthetic Aperture Radar

Simple meaning: What it is: Radar imaging. Instead of relying on sunlight like a normal optical camera, radar sends electromagnetic energy and measures the returned signal.

In your project: In your project: SAR backscatter patterns are the information the neural network receives. Your document describes the input as dual-polarized VV/VH SAR imagery.

3. VV and VH

Simple meaning: What they are: Two radar polarization channels. VV means vertical transmit/vertical receive; VH means vertical transmit/horizontal receive.

In your project: In your project: these are the two input channels instead of the usual three RGB channels. The network therefore receives a 2-channel image.

4. TIFF / .tif

Simple meaning: What it is: A raster image file format that can store numerical pixel data and, in geospatial workflows, location information.

In your project: In your project: rasterio reads the TIFF arrays and preserves the coordinate reference system and affine transform used later for mapping.

5. Raster

Simple meaning: What it is: Data represented as a grid of pixels/cells. Each cell contains a value.

In your project: In your project: the SAR image and segmentation masks are raster-like grids. The neural network works on these grids.

6. Segmentation

Simple meaning: What it is: Instead of giving one label to a whole image, the model assigns a class to individual pixels.

In your project: In your project: the model produces a map showing which pixels belong to sea, oil spill, look-alike, ship, or land.

7. Semantic Segmentation

Simple meaning: What it is: Pixel-wise classification where each pixel receives a semantic class.

In your project: In your project: this is the main ML task performed by the U-Net.

8. Dataset

Simple meaning: What it is: The collection of examples used by the ML system.

In your project: In your project: examples consist of Sentinel-1 imagery and corresponding masks/labels.

9. Train / Validation / Test Split

Simple meaning: What it is: Separating data so the model can learn, be monitored during development, and finally be tested on unseen data.

In your project: In your project: the document specifies 70% training, 15% validation, and 15% held-out test data. It says 180 test images were isolated before training.

10. Data Leakage

Simple meaning: What it is: Information from data that should be unseen accidentally influences training or model development.

In your project: In your project: the test set was isolated to prevent the final evaluation from influencing training.

11. Data Augmentation

Simple meaning: What it is: Creating varied training examples by applying transformations such as rotations and flips.

In your project: In your project: augmentation was applied only to the 70% training pool. The stated purpose is to make the model learn morphology rather than memorize orientation.

12. Mask / Ground Truth

Simple meaning: What it is: The correct answer map for an image, where pixels are assigned their known class.

In your project: In your project: the ground-truth mask is compared with the model's prediction during training and evaluation.

13. U-Net

Simple meaning: What it is: A neural-network architecture designed for segmentation. It has a contracting/feature-extraction side and an expanding/reconstruction side.

In your project: In your project: U-Net is the main segmentation architecture. It combines deeper semantic information with higher-resolution information through skip connections.

14. ResNet34

Simple meaning: What it is: A 34-layer residual neural network used here as the encoder/backbone.

In your project: In your project: ResNet34 extracts increasingly useful visual/spatial features. Residual connections help information and gradients pass through the network.

15. Encoder

Simple meaning: What it is: The part of the network that turns the input into increasingly abstract feature representations.

In your project: In your project: ResNet34 acts as the encoder.

16. Decoder

Simple meaning: What it is: The part that takes compressed/high-level features and reconstructs a detailed spatial prediction.

In your project: In your project: the U-Net decoder reconstructs a pixel-level segmentation from the encoder features.

17. Feature

Simple meaning: What it is: A learned pattern or signal that helps the network distinguish classes.

In your project: In your project: early layers can learn simpler patterns, while deeper layers combine them into more complex spatial patterns useful for distinguishing regions.

18. Convolution

Simple meaning: What it is: A sliding mathematical operation using learnable filters/kernels to combine nearby pixel values.

In your project: In your project: convolutional layers are used to extract spatial patterns from SAR data.

19. Kernel / Filter

Simple meaning: What it is: A small set of learnable numbers that slides over an input to detect a pattern.

In your project: In your project: kernels are part of the convolutional feature extraction performed by the network.

20. Skip Connection

Simple meaning: What it is: A shortcut that passes earlier feature information directly to a later part of the network.

In your project: In your project: U-Net uses skip connections to combine high-resolution spatial details with deeper semantic features.

21. Residual Connection

Simple meaning: What it is: A shortcut in ResNet where the input is added to a transformed version of itself: $F(x) + x$.

In your project: In your project: these connections help ResNet34 train deeper networks by improving information and gradient flow.

22. Input Channels = 2

Simple meaning: What it means: Each input image has two numerical channels rather than three RGB channels.

In your project: In your project: channel 1 is VV and channel 2 is VH.

23. Output Classes = 5

Simple meaning: What it means: The model predicts one of five semantic categories for each pixel.

In your project: In your project: Sea Surface, Oil Spill, Look-alike, Ship, and Land.

24. Class Imbalance

Simple meaning: What it is: Some classes can occupy far more pixels than others.

In your project: In your project: the document emphasizes that oil spills occupy a small fraction of ocean pixels, making ordinary accuracy potentially misleading.

25. Dice Loss

Simple meaning: What it is: A training loss based on overlap between prediction and ground truth. The Dice coefficient is $2|X \cap Y|/(|X|+|Y|)$; Dice loss is 1 minus the differentiable Dice score.

In your project: In your project: Dice loss was used to focus learning on overlap between predicted and actual spill regions.

26. Probability / Prediction

Simple meaning: What it is: The network's numerical confidence/score for possible classes before the final class decision.

In your project: In your project: these predictions are converted into a segmentation map used for evaluation and later post-processing.

27. Epoch

Simple meaning: What it is: One complete pass through the training dataset.

In your project: In your project: training ran for 20 epochs according to the documentation. The document tells us the number used, but not a rigorous reason why 20 was selected.

28. Batch Size

Simple meaning: What it is: The number of training images processed together before an update step.

In your project: In your project: the batch was reduced from 8/20 to 2 and sometimes 1 to avoid CUDA out-of-memory errors on the RTX 4060's 8 GB VRAM.

29. VRAM

Simple meaning: What it is: GPU memory used to hold model data, tensors, activations, gradients, and other computation state.

In your project: In your project: the 8 GB RTX 4060 constrained batch size and motivated memory optimization.

30. AMP — Automatic Mixed Precision

Simple meaning: What it is: A technique that uses lower precision for suitable operations while keeping numerical stability where needed.

In your project: In your project: torch.amp.autocast was used to reduce memory use during training. The document specifically describes FP16 casting.

31. Forward Pass

Simple meaning: What it is: Sending an input through the network to produce a prediction.

In your project: In your project: SAR tensors go through ResNet34/U-Net and produce class predictions.

32. Loss

Simple meaning: What it is: A number representing how different the model's prediction is from the ground truth.

In your project: In your project: Dice-based loss is used to guide training.

33. Backpropagation

Simple meaning: What it is: Computing how the loss changes with respect to the network's parameters so the parameters can be updated.

In your project: In your project: it is part of the training loop that allows the network to learn from its errors.

34. Gradient

Simple meaning: What it is: A mathematical direction indicating how changing a parameter would change the loss.

In your project: In your project: gradients are used to update the network's learned parameters during training.

35. Optimizer

Simple meaning: What it is: The algorithm that uses gradients to update model parameters.

In your project: In your project: the provided documentation does not name the optimizer, so this guide does not assume which one you used.

36. Weight / Parameter

Simple meaning: What it is: A learned numerical value inside the network.

In your project: In your project: training changes these values so predictions become more consistent with the training masks.

37. Overfitting

Simple meaning: What it is: When a model learns the training data too specifically and performs worse on unseen data.

In your project: In your project: validation data is used to monitor generalization. The documentation claims the losses mirrored each other, but that alone does not prove zero overfitting.

38. Checkpoint

Simple meaning: What it is: A saved copy of model parameters at a particular point in training.

In your project: In your project: best_oilspill_model.pth is described as the final checkpoint used for evaluation.

39. IoU — Intersection over Union

Simple meaning: What it is: Overlap divided by the union of prediction and ground truth: intersection / union.

In your project: In your project: the document reports 83.63% mean image-wise IoU and 65.71% global IoU for the oil class.

40. Dice Score

Simple meaning: What it is: Another overlap measure. Higher means prediction and ground truth overlap more.

In your project: In your project: the document reports 80.42% mean Dice for the oil class.

41. Inference

Simple meaning: What it is: Using a trained model to make predictions on an image without training it.

In your project: In your project: app_predict.py loads the trained model and predicts on TIFF imagery.

42. torch.no_grad()

Simple meaning: What it is: A PyTorch mode that disables gradient tracking when gradients are not needed.

In your project: In your project: it is used during inference to reduce unnecessary computation/memory use.

43. Normalization

Simple meaning: What it is: Scaling numerical input values into a range suitable for processing.

In your project: In your project: rasterio-ingested integer values are described as being normalized to float32 values from 0 to 1.

44. Connected Components

Simple meaning: What it is: A method for finding separate connected regions in a binary mask.

In your project: In your project: OpenCV connectedComponentsWithStats is used to identify predicted regions and remove very small clusters.

45. 50-pixel threshold

Simple meaning: What it is: A minimum region size used during the post-processing step.

In your project: In your project: components smaller than the stated 50-pixel threshold are removed. The documentation gives the value but does not establish how it was experimentally selected.

46. Vectorization

Simple meaning: What it is: Turning a raster mask made of pixels into geometric shapes/polygons.

In your project: In your project: rasterio.features.shapes converts the cleaned mask into geographic geometries.

47. CRS — Coordinate Reference System

Simple meaning: What it is: The system that tells software how numerical coordinates correspond to real places on Earth.

In your project: In your project: the original CRS is preserved during processing so predictions can be mapped correctly.

48. Affine Transform

Simple meaning: What it is: A mathematical mapping between raster row/column positions and real-world coordinates.

In your project: In your project: it is used to map predicted pixels back to their physical geographic positions.

49. EPSG:4326

Simple meaning: What it is: A commonly used geographic coordinate reference system using latitude and longitude (WGS 84).

In your project: In your project: final GeoJSON coordinates are bundled using EPSG:4326.

50. GeoJSON

Simple meaning: What it is: A JSON-based geographic data format that can store features such as polygons and their coordinates.

In your project: In your project: the final cleaned oil-spill regions are exported as GeoJSON so they can be displayed or integrated into mapping software.

The Entire Project in One Mental Model

Stage	What happens	Why it exists
1. Input	Sentinel-1 SAR TIFF → VV + VH	Give the model useful radar information
2. Dataset	Images + ground-truth masks	Provide examples and correct answers
3. Split	70% train / 15% validation / 15% test	Learn, monitor, then test on unseen data
4. Augment	Rotate/flip training examples	Increase variation in training data
5. Network	ResNet34 encoder + U-Net decoder	Extract features and reconstruct pixel-level prediction
6. Training	Forward → Dice loss → gradients → parameter updates	Make predictions progressively closer to masks
7. Evaluation	IoU + Dice on held-out test set	Measure performance on unseen images
8. Inference	Trained model predicts new TIFF	Use the learned model
9. Cleanup	Remove tiny connected regions	Reduce small predicted noise
10. Mapping	Pixels → geographic shapes → GeoJSON	Turn predictions into map-ready output

Key numbers to understand: 2 input channels (VV/VH), 5 output classes, 70/15/15 data split, 180 held-out test images, batch size 2/1, 20 epochs, 50-pixel post-processing threshold, and reported metrics of 83.63% mean image-wise IoU, 65.71% oil-class global IoU, and 80.42% oil-class mean Dice.

Study rule: For every number, ask: Is it a standard convention, a hardware constraint, an experimental choice, or simply the value used in this implementation? The supplied documentation does not establish the reason for every value.